Study guide: diff and patch

The diff command

You use the diff command to find the differences between two files. On its own, it's a bit hard to use; instead, use diff -u to find lines that differ in two files:

Using diff -u

You use the diff -u command to compare two files, line by line, and have the differing lines compared side-by-side in the same output. For an example of what this looks like, see below:

```
~$ cat menu1.txt
Menu1:

Apples
Bananas
Oranges
Pears

~$ cat menu2.txt
Menu:

Apples
Bananas
Grapes
Strawberries

~$ diff -u menu1.txt menu2.txt
--- menu1.txt   2019-12-16 18:46:13.794879924 +0900
+++ menu2.txt   2019-12-16 18:46:42.090995670 +0900
@@ -1,6 +1,6 @@
-Menu1:
+Menu:

 Apples
 Bananas
-Oranges
-Pears
```

```
+Grapes
+Strawberries
```

The patch command

The patch command is useful for applying file differences. See the example below, which compares two files. The comparison is saved as a .diff file, which is then patched to the original file!

```
~$ cat hello_world.txt
Hello World
~$ cat hello_world_long.txt
Hello World

It's a wonderful day!
~$ diff -u hello_world.txt hello_world_long.txt
--- hello_world.txt     2019-12-16 19:24:12.556102821 +0900
+++ hello_world_long.txt        2019-12-16 19:24:38.944207773 +0900
@@ -1 +1,3 @@
 Hello World
+
+It's a wonderful day!
~$ diff -u hello_world.txt hello_world_long.txt > hello_world.diff
~$ patch hello_world.txt < hello_world.diff
patching file hello_world.txt
~$ cat hello_world.txt
Hello World

It's a wonderful day!
```

Resources for more information

There are other interesting patch and diff commands such as patch -p1 and diff -r. For more information on these commands, check out the following resources:

- http://man7.org/linux/man-pages/man1/diff.1.html
- http://man7.org/linux/man-pages/man1/patch.1.html